

A Packing of 97 Lines in $\mathbb{R}^{11}$

Limin Ge*

October 8, 2026

Abstract

We give 97 1-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 0.826575. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.825051, obtained by deleting 1 of the 98 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 97 subspaces of dimension 1 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.826575 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 1$. The cell $N = 97$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.825051, obtained by deleting 1 of the 98 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(11 - 1)/11 \cdot 97/96 = 0.918561$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n1N97.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 80 seconds on one laptop, with about ten such searches running at once.

A Exact value

```
numerator:
275525242811364730208973946755457361804253918563
denominator:
333333333332948064411155399540977153059794767763
```

B Coordinates

97 blocks of 11 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m11n1N97.txt`.

-442894998821	423401318490	-258210949178	-513678946060
280011442100	-371239278869	-498235433413	-704410708218
-262088860964	166031865112	-93878029298	25413036511
1300417076	280105876865	185158235099	152901953594
114343920558	-296701956617	122265077058	166611593915
287859440364	-442699656854	243656921287	34525766909
-197541590936	-212231209182	40745166888	-134039132360
-447666239418	76868760289	-297470577979	-252161426480
385143499390	-617158786239	-194922510097	-256098024356
412938671149	166888131427	366492906440	150625435983
50260892293	-384077750192	658551405191	-523817800941
285361816367	136787257427	282170514626	-69824039529
281231045289	-318088124748	96810105712	22767707430
142855079254	17174459551	325969148301	-728154729076
356079508052	-114476916754	-340900805438	-6050056245
-249952160109	531570737338	314550665312	149745275902
557444259445	98840745422	104271853336	176109381149
281303632487	110504219270	65191724570	-119192483145
155571371538	-203384508643	-481155525407	190618125394
327002891505	190295561489	-50448466890	-289061674679
151187608906	96920989709	249383315229	-49379170141
-293146019996	160394590663	335354255842	10024077472
183200543295	-498362328105	-78322452850	520524084278
200846710919	-112937208262	-594590116512	212166539770
-76987633461	263100761608	-36604971135	-416861583225
-556069929675	400938028422	-347488781325	-112440188022
-301440965427	-343190021686	-301869182910	219883511296
473025866635	468576368080	-93915263261	-93612402669
-78281126029	-242622953934	160849065909	223228925182
20132494065	-159456303698	-459140934566	-67619748529
167405614835	194013566376	-5957883779	13584511360
444149047983	-118445652159	416235410807	621146533966
-253989965155	-35871284865	-308504478716	-34681320719
226061696389	78704913120	400038114921	-144655684208
137557472270	397797973004	-334663742458	-385237485796
37864254713	-786183524858	-46161736420	749562039142
-132197903019	-57202533451	-239799604012	57141509360
-396749589913	-99264758034	-376593404041	367726492604
-439839880318	-202606986160	-240096087185	-51931429719
-602052896991	291408413373	-129646732285	-266228268677
-282402093021	-639933057592	-444045528061	221725890047
-28218933545	-310062359266	-160056040523	18256392649
-74127579210	-82583511418	388609950601	77864060733
334192859526	342675574071	149644433337	215498265357
-174431333059	-151145935820	-540227226848	-212273167850
-90341011289	85431783095	-192916295831	366790915982
429264143797	271326233556	-152055918928	-345711413599
-224467283422	315185040561	-111089661118	-428468149272
-371388976794	156650002372	406737782044	508665740550
225693430713	299655754908	194505637256	326004964687
339498492816	229487291074	-20722892911	-46302238267
86793266045	-220326234023	271140151920	-166277329128
22022869027	-203019071562	324343406761	20376945256
-82791525528	569908213789	-462713875011	275020630864
-638560809465	-65407672183	125962836296	-188246491059
372021816511	169839244939	-1132303772	-192854326323
-284038946452	154360304212	-58979840518	637248033879
-77390883310	530263815293	-609750182606	-167286729861
47058456319	-402175684048	320359663932	446144704971
-67362096947	260170239574	548565912434	8419797870
-160276983533	73640013893	-34369176588	73152258798
-351913608141	111611822701	39617189891	500311249972
-198363456388	93397264481	4384359313	92739399326
-760436145605	-326767279308	341976598632	136475347862
-1541142601	332067668306	-164494533763	106281952398
-32166429143	-104427260074	277659363456	417384478458
72266895750	72328822762	216191606488	-272169609175
479393244761	-241453168919	-569812977774	169255274051
-49236249233	136745741848	33887057709	185190742632
-101809232712	-597805229247	130908818746	272229260200

37526704078	357472379595	304308207807	-94743380339
-24695950385	-403833058693	169017392444	-607971697200
529660516093	-511581460717	27379334061	-32207502874
59562255882	26406402528	-387313859832	-155546427669
207128006652	-25029743371	365662923241	-406277222567
-534466175418	247430474536	-255683230340	-381418951610
146675257986	126374522549	209234338951	214977891774
-70377513024	217713178053	299842495412	-70310847488
75232536782	-753896806232	363495935956	-607345305431
432958807867	10866084430	509727508192	453205925490
-345010898320	351349106728	-57868258443	-341439860139
-61825617963	-264904740382	-218182353271	-58156001524
-373555933089	326257822786	-838755552536	-348749767637
417335943014	-74441096405	200497573639	-211093782043
9837514515	215223404809	-121573997	-186714744972
-387861969810	296127629813	-7674547277	-59345570041
439679627303	-25341097240	355873844033	-88955625156
27474096463	153038291486	115840795349	42800632526
288368425881	-298854863743	-49257502326	-302417749410
-151957206440	-495958449323	-84363891204	551909074334
64432646450	50482735670	-206182201420	-535247056555
-256761140188	366345121029	-128481293303	49847883762
-460049216578	276524417611	35713273651	173190216665
78187860724	-128252893834	-430821683693	-448046734554
175380002005	-526273310516	671715189183	-146176649424
-595899296479	-82604660172	373802041739	-109825915681
-147233237547	-486520457287	-329335313169	-212421429165
444555527718	-109011951153	-102340299401	192131048577
-153323412512	-378349495625	105153608536	-245106739201
264183792246	-303470787527	170428235846	11728662292
280588134588	130900448864	56388554482	20168686401
310451890695	-284237053510	233597488350	405219368537
-141288367935	-11157962948	-73499463776	-59223926712
-403831064911	107058995985	-350812490672	462644976481
-480060367547	-342093947462	499333086446	-417497962446
101688756560	531796786183	158261395296	76512493521
-372331713037	397435310601	-275311878415	-556889068007
-318652017397	-465096138411	111208226669	-38742461059
-259958214060	-72150292760	-300606688542	318643385762
-265463850908	-529058931148	-367947386533	84878557759
-68427406027	190424149623	3036216023	-131603724000
-178219929558	265473791040	304094589683	-241676701852
-472832222143	-330965170189	-423698813804	399467085819
434990713818	223522900014	128970587726	223238593752
115086097497	-171505661535	-143979346327	544216390669
-293019145797	-140811951090	56215488519	253879955164
222757205315	-158560461161	336143382707	-32619049217
348798295323	-3894722399	81035515443	-8778094661
182206973754	193288325940	134710496610	579905286199
420370818450	100337446783	378844509078	111777097424
84998997828	-234379999707	151249303740	-145220193704
-46046531529	31262313089	30880902019	23285519849
329772034592	345293334447	12003966534	-177519546310
413976913096	70610730680	818220221327	-38632923211
44700610776	786499643134	-37711409855	243518596284
-101656501480	-320770500230	206465357071	230040599465
266507762777	-115868024535	76053008767	22576502161
-480508698987	-193730149449	560629750030	97530536640
-458940870609	-110671235525	266920824299	-7798055812
218025166479	-158446062923	331709868577	-72335501663
371644353749	213481492228	105735745775	428247239480
497540130867	426561704396	303863526808	-3308724354
53248612350	77406873982	29219452796	809970702867
-78012059821	-118719279177	94490195724	-18847281476
55283089324	-251834442520	-459930327377	-283887646113
-129296755635	19548991807	-362178058147	75507936328
199262858190	690086045194	-27433494828	64022958922
3146580017	-412897179928	-27088496524	41738201320
-324229692368	-67274183155	-510399501044	619616296529
-282077397042	124860191325	529856323487	-83873196898
-424356493333	213082654245	-9591959309	531798423174
565041660951	544343276066	-219633520080	397613412591
353166045993	212404179220	363317030105	-170890844415
-28763220548	-472055826202	443265382402	215502625619
380338383742	197625043757	-215149989664	-45299760869
195262199299	3225024000	-80285153455	198609033889
75099489855	-330206755316	-166459202716	-160930141553
-417781420449	263077354439	-69694868581	-557356704589
-389114255160	-368731295765	-564067804414	157419936784
-272760880750	146522529433	194866289407	-319766340002
226139867375	292967864582	-225989350971	-537815054793
224054204117	252463763132	-276188694320	-163337860790
-429339763118	-11781902570	-428341205891	91824150910
-412980932748	-437290541863	-352400077727	307336920410
117364692269	331251944947	119663100710	-275887442174
-580413750566	-169303372992	-139658066638	87851150346
191958737127	230108905852	-56507830745	-759686610784
-119664134205	-471020384490	-408616095275	-111138574758
-45087439599	-45837362740	188276371147	287991875438
-174426912984	169643643679	238953162262	116806456621
175645577788	463668225608	191385798828	-241633982692
413053444251	-128196889417	-510627081759	-161471175159
-294235245319	-135693704624	564225524086	381631109664
327304098660	556987172179	482213595821	-49829851049
-61343854532	-237820323456	164244802308	152411516532
89278539	-94592860169	61585794054	224519636252
63590881703	221268428673	112770146321	-102399709807
-248971516938	-370246536685	-123807345672	-258932489182
-294898465730	223050680743	-19746640491	167309637399
-47708076131	-35197022626	-200192771295	-283557612431
-316226503042	-585748381162	-2010430648	667738050705
-434030768262	-96322025436	440431665100	-241723814156

-132370274917	-171382636407	253425167123	165594334636
-33487366377	18365825405	-123081475377	-673786286972
55454170170	175835148137	156692895141	368567179509
-221322989557	152829346802	178070580072	197639674961
-488941714695	395277020558	545316385850	372086346462
577046408284	250591116075	235354889352	-87360410047
-266036818258	-223647983436	165050122624	14813274115
-240092964388	25984389805	660005947764	-187135558980
54524943998	-61880789291	-162058505420	176041113613
354485086892	-348871272101	83927382872	560950747370
-579255657913	155145305271	-255607657401	474960602813
168974659498	392755777624	-18529340024	-180895482359
-175482336180	-11386494836	170708964926	275465825755
-79502362735	527328616097	-520369514717	48010294898
-92306915614	-350330236581	-106862814904	392047081067
15484853346	-168317966771	-334273846766	346886031948
228665168791	-386461963254	-128872117404	92610164385
-296102683764	34927350543	206915714180	-306842831393
35523306730	331607746988	75232158416	283062739982
-309931934440	100786248855	-205065690449	299335513750
-52672303014	478535827636	250322825605	358644514996
-185869869844	251305925025	-358602040269	-103940711431
-182038471088	-461746421721	-489657699750	-40065178059
70644492939	41768192044	-260578872140	-407530945859
-236217206786	164271899282	-301035299146	111580561256
448716855801	55447706955	186352226259	-269742471679
552440874874	-147970802476	118113077693	403065566224
-668508973050	-98066880685	-454581282963	425830821990
64689205401	384961638171	-316849554133	-348813494283
238998023585	199041821109	-170507365428	10760737414
5360442821	-493466481020	403156025411	446024878412
150799553553	463589393522	-109085178134	537688223571
-329736864748	-5271847563	177162953431	38649217155
380841763758	24362432751	508562522293	196277317673
-146846442402	192211816449	245219177087	-100221712975
-218414995333	170780160754	97984484291	-428309059419
-206798189075	333917942126	339621864784	12623238775
-321279375072	317473210259	69843126399	84552036311
435549429915	-675192877256	-65901306547	-386453816456
233784612884	75266906632	324017499014	409590417331
-272353100964	-6878141965	-463292287502	354478670543
444969225056	-210864978467	299875043871	541939309791
-157999767011	-347172226192	-359712321299	183725082246
-249695736155	98198959355	491760320123	182377794547
257394363176	-52020340215	-164103183236	-340602902060
-30910800732	140729765288	-241105451486	-22048486550
418187840656	-279496645613	156579415380	73371485289
-278192412241	469309229129	-369159939099	-56029004299
276904884211	524303004912	-293892951033	-197765028740
-87016519924	-398312157819	216095428549	426304676790
-43819968902	-270356509204	-240063015348	129540972445
55123730119	191577093295	-25082340730	32159504759
-236633540713	-74881632485	162419165994	212607673043
-863668329711	-310978974935	300318767764	-255864347245
-81111436660	-427859637288	679669915007	56061507345
-160188213342	-9310212453	205761603585	-9496820089
204387211038	-37903169795	-164358666804	471501562083
331263399605	100857177651	-539864576376	53926758861
23876305074	-296833240788	2394078853	301862395016
-35098644975	325551215360	-426404861387	-628257842265
151766881828	508070736060	337507457328	396525652923
-61446188613	266636257296	158364390737	51177338627
-23823794434	-401852462941	-74158463271	373931492193
-99824694906	-154022635125	3263069411	326670240574
100538147230	63260746288	-197173738605	-2990033139
525381520812	-86695625990	-524739478781	-499474919839
301170256955	54713954372	151105062377	68152831277
-95489711110	-117242899717	212501691243	-264248905051
-543915156862	473985794016	-63237692529	288717239353
-475728550651	92803204765	424932171995	-281764135752
-233541890804	-155173156483	-466252927947	42098745363
187904193265	387590320649	-306925520873	-55429576991
230049422360	-313443746772	-132717093456	-292220891475
-297229786520	-463288085604	4376005508	230510754401
-424286538413	501278033157	557654457034	-136518768767
-194733409723	-71195958355	281341382202	-105630823667
-65534990629	-501167778167	76290055860	323906258347
-100749943219	4997526861	104339116057	-549696976551
-247009191236	-456262371096	-281722041013	352164395304
688889320783	-228635096589	-445250893694	206943970607
-22394489758	55285339492	-277899879705	173962968529
234424034221	220126725664	14333292281	-9969544497
-24033275513	527556302498	346521374264	476850493135
417216773804	8840960495	540681382295	-535686111617
440270064229	387727466023	266512658437	152194416328
-78049378087	-55217866552	-418694031498	205859430218
27401928870	137837992081	194493277621	138536019678
54751595084	56826177224	45087266128	371325842746
683980733813	-416827163235	145274710671	403787360945
144823510026	47117319148	-67863893104	78485139910
-345923815460	327009572718	27959577959	-112984925696
-16017534925	738196754099	-150997698510	77635871574
113803281007	-151376296262	382098897685	-516245175915
751721582332	52784899052	-89836191424	189520132228
278150806144	-211807567817	-141262544824	